

movie theatre management system
use case, abuse case, and Security Use case cards

Use case card: login.

use case id:	F001
name:	login
created by/author:	Project group
date created:	14 Feb 2026
actor:	Staff / Manager / Maintenance Staff
description/summary:	A system user shall be able to log into the Movie Theatre Management System so they can access the features that match their assigned role
preconditions:	System is running on the theatre PC User account exists in the database Database connection is active
postconditions:	If the credentials are correct, access is granted based on role. The authenticated user is redirected to the correct interface Failed attempts are reset after a successful login. Locked accounts are denied access
normal course of events:	1 User enters username and password 2 User clicks Login 3 System checks that the fields are not empty 4 System checks whether the account is locked 5 System retrieves the user record from the users table using the entered username 6 System verifies the password using BCrypt 7 If authentication succeeds, the system resets failed attempts 8 System loads the interface based on the user role
exceptions:	Empty fields → access denied. Invalid credentials → access denied Locked account → access denied After the third failed login attempt → account is locked and access is denied
acceptance criteria:	Valid credentials grant access according to the assigned role. Locked accounts cannot log in. Account is locked after 3 failed attempts Failed attempts are reset after a successful login

Abuse case card: sql injection in login form.

abuse case id:	A001
name:	sql injection in login form
created by/author:	Project group
date:	14 Feb 2026
priority:	High
scope:	Authentication and database
development environment:	Java Desktop Application (JavaFX) with relational database
mis-actors:	Malicious insider / staff
access right levels:	Low level system users
point of entry:	Login form input fields
security attributes affected:	Confidentiality, Integrity, Availability
description:	An attacker tries to enter SQL code in the login fields to bypass authentication
sophistication:	Medium
preconditions:	Access to the login form Unsafe query handling would allow input to change SQL execution
assumptions:	If the login query is built unsafely, SQL code could be injected through the username or password field
postconditions:	Unauthorized access is granted
related use cases:	F001 - login
related threats:	SQL Injection attack
exceptions:	N/A
acceptance criteria:	SQL injection attempts in the login form are rejected and do not affect authentication

Security Use case card: confirm and secure login input.

use case id:	S001
name:	validate and secure login input
created by/author:	Project group
date created:	14 Feb 2026
actor:	System
description/summary:	The system checks the login fields, runs the authentication query safely using a prepared statement, and verifies passwords with BCrypt
preconditions:	User submits login credentials
postconditions:	Query is executed safely Malicious input is treated as plain text Only correct credentials are authenticated
normal course of events:	1 User enters username and password 2 System checks that all required fields are filled in 3 System runs the authentication query using a prepared statement 4 System compares the entered password against the stored BCrypt hash 5 Access is only granted if the credentials are correct
exceptions:	Empty fields → deny login Invalid credentials → deny login
acceptance criteria:	SQL injection attempts fail Only valid credentials allow login

Abuse case card: brute force login attempt.

abuse case id:	A002
name:	brute force login attempt
created by/author:	Project group
date:	14 Feb 2026
priority:	High
scope:	Authentication module
development environment:	Java Desktop Application
mis-actors:	Malicious insider
access right levels:	Low level system users
point of entry:	Login screen
security attributes affected:	Confidentiality, Availability
description:	An attacker keeps trying many password combinations to guess a valid account
sophistication:	Low
preconditions:	The account lockout feature is missing, turned off, or not working properly
assumptions:	The attacker has enough time to keep guessing passwords
postconditions:	Unauthorized access is gained after repeated attempts
related use cases:	F001 - login
related threats:	Credential guessing attack
exceptions:	N/A
acceptance criteria:	The system allows repeated failed login tries without locking the account

Security Use case card: implement account lockout mechanism.

use case id:	S002
name:	implement account lockout mechanism
created by/author:	Project group
date created:	14 Feb 2026
actor:	System
description/summary:	The system tracks failed login attempts, locks the account once the limit is reached, and stores the lock action in the actions log
preconditions:	User attempts login
postconditions:	Account is locked after 3 failed login attempts. The lock action is stored in the actions log
normal course of events:	1 User attempts to log in 2 System counts failed login attempts. 3 After the third failed attempt, the account is locked. 4 System stores the account lock action in the actions log. 5 If the user logs in successfully before reaching the limit, the failed-attempt counter is reset
exceptions:	Successful login resets the counter
acceptance criteria:	Lockout is enforced after 3 failed attempts Locked account cannot authenticate. Account lock action is logged

Use case card: book ticket.

use case id:	F002
name:	book ticket
created by/author:	Project group
date created:	14 Feb 2026
actor:	Staff
description/summary:	A staff member shall be able to book a movie ticket by choosing a showtime and an available seat, then selecting the payment method so the booking is saved in the system
preconditions:	Staff user is logged in At least one valid showtime exists in the system Database connection is active
postconditions:	Ticket record is stored Payment record is stored Booking action is stored in the actions log
normal course of events:	1 Staff opens the showtimes screen 2 System displays all scheduled showtimes 3 Staff enters a showtime number 4 Staff enters a seat number 5 System checks that the showtime exists 6 System checks that the selected seat is not already taken 7 System opens the booking screen 8 Staff sees the fixed ticket price 9 Staff chooses a payment method 10 System saves the ticket record 11 System saves the matching payment record 12 System commits the booking, shows a success message, and stores the booking action in the actions log
exceptions:	Showtime not found → show an error message Seat is already taken → show an error message Payment method not selected → show an error message If saving ticket or payment fails → the whole operation is cancelled and incomplete booking data is not saved
acceptance criteria:	Seat cannot be double booked for the same showtime Booking cannot continue unless a payment method is selected Ticket and payment are stored successfully Booking action is logged

Abuse case card: double booking of seat.

abuse case id:	A003
name:	double booking of seat
created by/author:	Project group
date:	14 Feb 2026
priority:	High
scope:	Seat reservation module
development environment:	Java Desktop Application with relational database
mis-actors:	Malicious insider
access right levels:	Low level system users
point of entry:	Showtimes and booking screens
security attributes affected:	Integrity, Availability
description:	An attacker or user tries to reserve the same seat for the same showtime before the booking process finishes
sophistication:	Medium
preconditions:	Seat availability checks are bypassed, delayed, or not enforced correctly
assumptions:	The system does not fully prevent duplicate reservation attempts during booking
postconditions:	More than one booking exists for the same seat and showtime
related use cases:	F002 - book ticket
related threats:	Data tampering
exceptions:	N/A
acceptance criteria:	The same seat is stored more than once for the same showtime

Security Use case card: validate seat availability and save booking atomically.

use case id:	S003
name:	validate seat availability and save booking atomically
created by/author:	Project group
date created:	14 Feb 2026
actor:	System
description/summary:	The system must check that the showtime exists and the seat is free before saving the booking, and must save the ticket and payment together so either both are saved or neither is
preconditions:	Showtime selected Seat selected Booking transaction started
postconditions:	Seat cannot be reserved twice for the same showtime Ticket and payment records stay consistent
normal course of events:	1 System verifies that the showtime exists 2 System checks whether the selected seat is already taken for that showtime 3 If the seat is available, the booking process continues 4 System inserts the ticket record 5 System retrieves the generated ticket ID 6 System inserts the matching payment record linked to that ticket 7 System commits the transaction if all steps succeed
exceptions:	If the seat is already reserved, reject the booking If ticket or payment storage fails, the transaction is rolled back
acceptance criteria:	Duplicate booking is prevented Ticket and payment records stay consistent after booking

Abuse case card: ticket price manipulation.

abuse case id:	A004
name:	ticket price manipulation
created by/author:	Project group
date:	14 Feb 2026
priority:	High
scope:	Booking and payment module
development environment:	Java Desktop Application
mis-actors:	Malicious insider
access right levels:	Low level system users
point of entry:	Booking and payment screen
security attributes affected:	Integrity
description:	An attacker tries to change the ticket price during the booking or payment process
sophistication:	Low
preconditions:	The system trusts a changed value instead of the internal price
assumptions:	Price protection is not enforced during booking
postconditions:	Payment is stored with an altered amount
related use cases:	F002 - book ticket
related threats:	Data tampering
exceptions:	N/A
acceptance criteria:	Payment is processed with an altered price

Security case card: internal ticket price validation

use case id:	S004
name:	internal ticket price validation
created by/author:	Project group
date created:	14 Feb 2026
actor:	System
description/summary:	The system must set the ticket price internally and use that same value when saving the payment amount
preconditions:	Booking confirmation started
postconditions:	Payment matches the fixed ticket price set by the system. Staff users do not enter the ticket price manually
normal course of events:	1. System sets the ticket price internally 2. Staff selects the payment method 3. System stores the ticket using the internal price 4. System stores the payment amount using the same internal price value
exceptions:	Manual price entry is not allowed because the price is assigned by the system
acceptance criteria:	The ticket price is assigned by the system. The same internal price value is used for both ticket and payment storage

Use case card: manage movies.

use case id:	F003
name:	manage movies
created by/author:	Project group
date created:	14 Feb 2026
actor:	Manager
description/summary:	The manager should be able to add, update, delete, and view movies so the theatre keeps an updated movie list for scheduling
preconditions:	Manager is logged in Database connection is active
postconditions:	Movie record is added, updated, deleted, or displayed Valid add, update, and delete actions are stored in the actions log
normal course of events:	1. Manager opens Manage Movies 2. Manager chooses Add, Update, delete, or Show All Movies 3. System displays the related form or movie list 4. Manager enters or selects movie details 5. System validates input where needed 6. System updates or retrieves data from the database 7. System confirms the action or displays the movie list
exceptions:	Invalid input → validation error Attempt to delete a movie that does not exist → action denied Movie already linked to a showtime → deletion is blocked
acceptance criteria:	Movie details are stored correctly. Invalid data is rejected. A movie with a scheduled showtime cannot be deleted Valid add, update, and delete actions are logged

Abuse case card: unauthorized or invalid movie change.

abuse case id:	A005
name:	unauthorized or invalid movie change
created by/author:	Project group
date:	14 Feb 2026
priority:	High
scope:	Movie management module
development environment:	Java Desktop Application with relational database
mis-actors:	Unauthorized user / malicious manager
access right levels:	Low level system users / Manager
point of entry:	Manage Movies screen and add/update movie forms
security attributes affected:	Integrity, Availability
description:	A non-manager tries to reach movie management, or a manager enters invalid movie data to damage or corrupt records
sophistication:	Low
preconditions:	Role checks or movie input validation are missing or not enforced
assumptions:	The system may allow bad movie changes if access control or validation is weak
postconditions:	Movie data is changed by the wrong user or invalid movie values are stored
related use cases:	F003 - manage movies
related threats:	Privilege escalation, Input manipulation
exceptions:	N/A
acceptance criteria:	A non-manager can change movie data, or invalid movie input is accepted

Security Use case card: restrict and validate movie management.

use case id:	S005
name:	restrict and validate movie management
created by/author:	Project group
date created:	14 Feb 2026
actor:	System
description/summary:	The system must provide movie management through the Manager interface, confirm movie fields before saving changes, and deny movie deletion if the movie already has a scheduled showtime
preconditions:	A user logs in and navigates through the system, or submits movie details
postconditions:	Movie management options appear in the Manager interface Invalid movie input is rejected Movie deletion is blocked when the selected movie is already linked to a showtime Valid actions are logged
normal course of events:	1. User logs in 2. System loads the interface based on the user role 3. If the role is Manager, the system shows movie management options 4. Manager opens the required movie action 5. System validates movie name, director, duration, description, and genre 6. If deleting a movie, the system checks whether the selected movie is already linked to a showtime 7. If valid, the system saves the change and logs the action
exceptions:	User is not routed to the Manager interface → movie management is not available in the normal flow Validation failure → operation denied Selected movie is already linked to a showtime → deletion is blocked
acceptance criteria:	Movie management options appear in the Manager interface only Invalid movie input is rejected Movie deletion is blocked when the selected movie is already linked to a showtime Valid add, update, and delete actions are logged

Use case card: manage showtimes.

use case id:	F004
name:	manage showtimes
created by/author:	Project group
date created:	14 Feb 2026
actor:	Manager
description/summary:	The manager shall be able to add, delete, and view showtimes so the theatre can control when movies are scheduled
preconditions:	Manager is logged in Movie exists. Room exists in the system. Valid date and time slot are entered when adding a showtime. Database connection is active
postconditions:	Showtime record is added, deleted, or displayed Valid add and delete actions are stored in the actions log
normal course of events:	1. Manager opens Manage Showtimes 2. Manager chooses Add, Delete, or View All Showtimes 3. If adding, the manager enters movie name, room number, date, and time slot 4. System validates the input. 5. System checks whether the movie exists. 6. System checks whether the room exists in the system. 7. System checks for a showtime conflict in the same room, date, and time slot. 8. System saves the valid change or displays the showtime list
exceptions:	Invalid input → validation error Movie not found → action denied Conflicting showtime → action denied Attempt to delete a showtime with booked tickets → action denied
acceptance criteria:	Conflicting showtimes are not stored Showtimes with linked tickets cannot be deleted Valid add and delete actions are logged

Abuse case card: unsafe showtime management.

abuse case id:	A006
name:	unsafe showtime management
created by/author:	Project group
date:	14 Feb 2026
priority:	High
scope:	Showtime management module
development environment:	Java Desktop Application with relational database
mis-actors:	Malicious manager
access right levels:	Manager
point of entry:	Manage Showtimes screen
security attributes affected:	Integrity, Availability
description:	A manager tries to create a conflicting showtime or delete a showtime that already has booked tickets
sophistication:	Low
preconditions:	Conflict checks or deletes checks are missing or not enforced correctly
assumptions:	The system may allow invalid showtime actions if the checks fail
postconditions:	Overlapping showtimes are stored, or a showtime with tickets is deleted
related use cases:	F004 - manage showtimes
related threats:	Data tampering
exceptions:	N/A
acceptance criteria:	The system allows a conflicting showtime to be stored or allows deletion of a showtime that already has booked tickets

Security Use case card: validate and protect showtime management

use case id:	S006
name:	validate and protect showtime management
created by/author:	Project group
date created:	14 Feb 2026
actor:	System
description/summary:	The system must validate showtime input, check that the selected room exists, reject schedule conflicts, and block deletion of showtimes that already have tickets.
preconditions:	Manager tries to add or delete a showtime
postconditions:	Conflicting showtimes are rejected Showtimes with existing tickets cannot be deleted Valid deletions are logged
normal course of events:	1. Manager enters showtime data or selects a showtime to delete. 2. System validates room number, date, and time slot. 3. System checks whether the movie exists. 4. System checks whether the room exists in the system. 5. System checks for an existing showtime in the same room, date, and time slot. 6. If deleting, system checks whether tickets already exist for that showtime. 7. If all checks pass, the system saves the change and logs the valid action.
exceptions:	Validation failure → operation denied Conflict detected → operation denied Tickets already exist for the showtime → deletion denied
acceptance criteria:	Overlapping showtimes cannot be stored Showtimes with linked tickets cannot be deleted Valid add and delete actions are logged

Use case card: view sold tickets.

use case id:	F005
name:	view sold tickets
created by/author:	Project group
date created:	14 Feb 2026
actor:	Manager
description/summary:	The manager shall be able to view all sold tickets so the theatre can review booking records
preconditions:	Manager is logged in Database connection is active
postconditions:	The sold ticket list is displayed with ticket details
normal course of events:	1. Manager opens the Show All Tickets screen 2. System retrieves ticket records together with movie name and show date 3. System displays the sold ticket list
exceptions:	No ticket data available → empty list is shown Database failure → error is displayed
acceptance criteria:	The manager can open the sold ticket screen Ticket details are displayed correctly

Abuse case card: unauthorized access to sold tickets.

abuse case id:	A007
name:	unauthorized access to sold tickets
created by/author:	Project group
date:	14 Feb 2026
priority:	Medium
scope:	Ticket viewing module
development environment:	Java Desktop Application
mis-actors:	Unauthorized internal user
access right levels:	Low level system users
point of entry:	Show All Tickets screen
security attributes affected:	Confidentiality
description:	A user who is not supposed to review sold tickets tries to open the sold ticket screen
sophistication:	Low
preconditions:	Access control to the ticket screen sold is not enforced
assumptions:	The system may expose ticket data if screen access is not restricted
postconditions:	Ticket details are shown to an unauthorized user
related use cases:	F005 - view sold tickets
related threats:	Privilege escalation
exceptions:	N/A
acceptance criteria:	A user who is not the Manager can open and view the sold ticket list

Security Use case card: restrict sold ticket access to manager.

use case id:	S007
name:	restrict sold ticket access to manager
created by/author:	Project group
date created:	14 Feb 2026
actor:	System
description/summary:	The system must make the sold ticket screen available through the Manager interface only
preconditions:	A user logs in and navigates through the system
postconditions:	Sold ticket viewing is available through the Manager interface only. Users in other roles are not routed to it in the normal flow
normal course of events:	1. User logs in 2. System loads the interface based on the user role 3. If the role is Manager, the system shows the Show All Tickets option and opens the sold ticket screen when selected 4. If the role is not Manager, this option is not provided in the normal flow
exceptions:	System error → do not open the sold ticket screen
acceptance criteria:	The Show All Tickets option appears in the Manager interface only. Users in other roles are not routed to the sold ticket screen in the normal flow

Use case card: generate reports.

use case id:	F006
name:	generate reports
created by/author:	Project group
date created:	14 Feb 2026
actor:	Manager
description/summary:	The manager shall be able to generate Movie Popularity, Peak Hours, Sales Summary, Show All Actions, and Maintenance reports to watch theatre operations
preconditions:	Manager is logged in Database connection is active
postconditions:	The selected report is generated and displayed
normal course of events:	1. Manager opens the Reports module 2. Manager chooses one of the available report types 3. System retrieves the needed data from the database 4. System displays the selected report
exceptions:	No data available → empty report or message is shown Database failure → error is displayed
acceptance criteria:	Reports are generated accurately Only the Manager can access the reports module

Abuse case card: unauthorized report access.

abuse case id:	A008
name:	unauthorized report access
created by/author:	Project group
date:	14 Feb 2026
priority:	High
scope:	Reports module
development environment:	Java Desktop Application
mis-actors:	Unauthorized internal user
access right levels:	Low level system users
point of entry:	Reports menu
security attributes affected:	Confidentiality
description:	A non-manager user tries to access theatre reports that should be limited to the Manager
sophistication:	Low
preconditions:	Role-based access control is not enforced for the report's module
assumptions:	The system may display sensitive report data if it does not verify the user role
postconditions:	Restricted reports are shown to an unauthorized user
related use cases:	F006 - generate reports
related threats:	Privilege escalation
exceptions:	N/A
acceptance criteria:	A non-manager can open and view restricted reports

Security Use case card: restrict report access to manager.

use case id:	S008
name:	restrict report access to manager
created by/author:	Project group
date created:	14 Feb 2026
actor:	System
description/summary:	The system must provide the Reports module through the Manager interface only
preconditions:	A user logs in and navigates through the system
postconditions:	Reports are available through the Manager interface only. Users in other roles are not routed to the Reports module in the normal flow
normal course of events:	1. User logs in 2. System loads the interface based on the user role 3. If the role is Manager, the system shows the Reports module 4. If the role is not Manager, the Reports module is not provided in the normal flow
exceptions:	System errors → do not open the Reports module
acceptance criteria:	The Reports module appears in the Manager interface only. Users in other roles are not routed to it in the normal flow

Use case card: write maintenance report.

use case id:	F007
name:	write maintenance report
created by/author:	Project group
date created:	14 Feb 2026
actor:	Maintenance Staff
description/summary:	Maintenance staff shall be able to write a maintenance report so the theatre can record equipment problems
preconditions:	Maintenance Staff is logged in Database connection is active
postconditions:	Maintenance report is stored Action record is stored in the actions log Confirmation is displayed
normal course of events:	1. Maintenance Staff opens the maintenance module 2. User enters equipment name 3. User enters report details 4. System validates the input 5. System stores the maintenance report 6. System stores the related action log entry 7. System shows confirmation
exceptions:	Invalid input → validation error Database failure → error is displayed and the report is not saved
acceptance criteria:	Maintenance report is stored correctly Invalid input is rejected The action is logged

Abuse case card: unauthorized or invalid maintenance report submission.

abuse case id:	A009
name:	unauthorized or invalid maintenance report submission
created by/author:	Project group
date:	14 Feb 2026
priority:	High
scope:	Maintenance reporting module
development environment:	Java Desktop Application
mis-actors:	Unauthorized internal user / malicious maintenance user
access right levels:	Low level system users / Maintenance Staff
point of entry:	Maintenance report screen
security attributes affected:	Integrity, Availability
description:	A user who should not use the maintenance module tries to submit a report, or a maintenance user enters invalid report data
sophistication:	Low
preconditions:	Role checks or maintenance input validation are missing or not enforced
assumptions:	The system may accept fake or invalid maintenance reports if access control or validation is weak
postconditions:	An unauthorized or invalid maintenance report is stored
related use cases:	F007 - write maintenance report
related threats:	Privilege escalation, Input manipulation
exceptions:	N/A
acceptance criteria:	A non-maintenance user can submit a report, or invalid report input is accepted

Security Use case card: restrict and validate maintenance report submission.

use case id:	S009
name:	restrict and validate maintenance report submission
created by/author:	Project group
date created:	14 Feb 2026
actor:	System
description/summary:	The system must provide maintenance report submission through the Maintenance Staff interface and validate the report fields before saving
preconditions:	A user logs in and navigates through the system, or submits maintenance report details
postconditions:	Maintenance report submission is available through the Maintenance Staff interface. Invalid maintenance input is rejected Valid report submissions are logged
normal course of events:	1. User logs in 2. System loads the interface based on the user role 3. If the role is Maintenance Staff, the system shows the maintenance reporting option 4. Maintenance Staff enters equipment name and report details 5. System validates the report fields 6. If valid, the system stores the report and logs the action
exceptions:	User is not routed to the Maintenance Staff interface → report submission is not available in the normal flow. Validation failure → operation denied
acceptance criteria:	Maintenance report submission appears in the Maintenance Staff interface only. Invalid maintenance input is rejected. Valid submissions are logged

Use case card: view maintenance reports.

use case id:	F008
name:	view maintenance reports
created by/author:	Project group
date created:	14 Feb 2026
actor:	Manager / Maintenance Staff
description/summary:	The Manager and Maintenance Staff shall be able to view maintenance reports so they can review reported equipment issues
preconditions:	Manager or Maintenance Staff is logged in Database connection is active
postconditions:	Maintenance report list is displayed
normal course of events:	1. User opens the maintenance reports screen 2. System retrieves maintenance report records from the database 3. System displays the maintenance report list
exceptions:	No report data available → empty list is shown Database failure → error is displayed
acceptance criteria:	Authorized users can open the maintenance report list Maintenance report details are displayed correctly

Abuse case card: unauthorized viewing of maintenance reports.

abuse case id:	A010
name:	unauthorized viewing of maintenance reports
created by/author:	Project group
date:	14 Feb 2026
priority:	Medium
scope:	Maintenance report viewing module
development environment:	Java Desktop Application
mis-actors:	Unauthorized internal user
access right levels:	Low level system users
point of entry:	Maintenance report list screen
security attributes affected:	Confidentiality
description:	A user who is not the Manager or Maintenance Staff tries to view maintenance reports
sophistication:	Low
preconditions:	Access control to the maintenance report list is not enforced
assumptions:	The system may show maintenance reports to unsupported roles if it does not check access
postconditions:	Maintenance report data is shown to an unauthorized user
related use cases:	F008 - view maintenance reports
related threats:	Privilege escalation
exceptions:	N/A
acceptance criteria:	A user who is not Manager or Maintenance Staff can open and view maintenance reports

Security Use case card: restrict maintenance report viewing by role.

use case id:	S010
name:	restrict maintenance report viewing by role
created by/author:	Project group
date created:	14 Feb 2026
actor:	System
description/summary:	The system must make maintenance report viewing available to the Manager and Maintenance Staff interfaces only
preconditions:	A user logs in and navigates through the system
postconditions:	Maintenance report viewing is available to authorized roles only. Users in other roles are not routed to this screen in the normal flow
normal course of events:	1. User logs in 2. System loads the interface based on the user role 3. If the role is Manager or Maintenance Staff, the system allows opening the maintenance report list 4. If the role is not allowed, this option is not provided in the normal flow
exceptions:	System errors do not open the maintenance report list
acceptance criteria:	Maintenance report viewing is available only through the Manager and Maintenance Staff interfaces. Users in other roles are not routed to the maintenance report list in the normal flow

